land, the possibility of rainfall from them reduces. So, most parts of India do not have rainfall in the winter season.

89. **Statement 1: INCORRECT**—Thunderstorms mostly occur on ground where the temperature is high. Thunderstorms are less frequent on water bodies due to low temperature.

Statement 2: CORRECT—In its cumulus stage, Ground is significantly heated due to solar insolation. Intense convection of moist hot air builds up a towering cumulonimbus cloud.

Statement 3: INCORRECT—The speed of isolated storms is typically about 20 km (12 miles) per hour, but some storms move much faster.

In extreme circumstances, a supercell storm may move 65 to 80 km (about 40 to 50 miles) per hour.

90. Tropical cyclones are violent storms that originate over oceans in tropical areas and move over to the coastal areas bringing about large scale destruction due to violent winds (squalls), very heavy rainfall (torrential rainfall) and storm surge.

They are irregular wind movements involving closed circulation of air around a low pressure center. This closed air circulation (whirling motion) is a result of rapid upward movement of hot air which is subjected to Coriolis force. The low pressure at the center is responsible for the wind speeds.

The cyclonic wind movements are anti-clockwise in the northern hemisphere and clockwise in the southern hemisphere (This is due to Coriolis force).

The cyclones are often characterised by existence of an anticyclone between two cyclones.

91. **Conditions Favourable for Tropical Cyclone Formation:**
    1. Large sea surface with temperature higher than 27°C,
    2. Presence of the Coriolis force enough to create a cyclonic vortex,
    3. Small variations in the vertical wind speed,
    4. A pre-existing weak low-pressure area or low- level-cyclonic circulation,
    5. Upper divergence above the sea level system.

92. **Polar or Arctic Cyclones:**